

Auditable AI Decision Systems with Bounded Decisions, Explainability, and Confidence Scores

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Abstract

Real-world AI systems must operate under ambiguity and incomplete information. Forcing a binary decision when evidence is insufficient produces overconfident and brittle outcomes. This work presents an auditable AI decision system with a bounded three-class decision space (YES, NO, TBD), structured explainability through auxiliary Human Values (10 labels) and Emotions/Sentiments (28 labels), and per-decision confidence scores. The auxiliary outputs are retained at inference time as control variables for downstream steering, thresholding, and reason-code generation. The system is trained on up to 181,000 labeled decision events and evaluated on a stratified held-out test split of 22,748 examples (YES: 5,871; NO: 3,159; TBD: 13,718; majority class TBD at 60.3%). It achieves Macro F1 = 0.8215 and Accuracy = 0.8485, outperforming a majority-action baseline (Macro F1 = 0.2508, Accuracy = 0.6030). Per-class F1 scores are 0.8302 (YES), 0.7524 (NO), and 0.8819 (TBD). Inference latency over 200 runs is $p_{50} = 200$ ms and $p_{95} = 415$ ms on an NVIDIA Tesla T4 GPU. These results demonstrate that bounded, defer-aware decision control provides a practical and auditable interface for AI systems operating under ambiguity and class imbalance. This work argues that bounded, defer-aware decision outputs, combined with structured auxiliary signals, form a more reliable and auditable interface for AI systems than traditional forced-decision classifiers.

1 Introduction

AI systems in production must handle inputs where evidence is incomplete, contradictory, or insufficient to support confident action. Traditional classifiers force a hard decision regardless of input quality, producing systems that are confident by construction but unreliable under ambiguity. A system that cannot express "not yet determinable" as an explicit output conflates low-confidence cases with high-confidence ones, degrading reliability and auditability.

Unlike prior approaches that treat abstention as a post-hoc confidence threshold, the TBD class is learned as a first-class outcome, allowing the model to internalize deferral behavior during training rather than relying on external calibration.

A second challenge is that decision outcomes are not determined by input content alone. The same input may require different handling under different execution-time conditions—varying risk tolerance, operational constraints, or deployment context. Systems that do not encode this context cannot adapt without retraining.

This work addresses both problems through a bounded decision formulation and structured explainability, and reframes classification as a decision interface problem rather than a label prediction task. The bounded decision space $\mathcal{Y}_d = \{\text{YES}, \text{NO}, \text{TBD}\}$ makes deferral an explicit, auditable output. Unlike prior approaches that treat abstention as a post-hoc confidence filter on a binary classifier

[3, 5, 4], TBD is learned as a first-class outcome, enabling the model to internalize when to defer under uncertainty.

In addition to decision prediction, the system separates decision control by retaining auxiliary Human Values and Emotions/Sentiments channels at inference time as structured control variables. These signals encode decision-policy context and allow downstream systems to apply thresholds, constraints, and reasoning logic without retraining the model. While multi-task learning [1] is typically used for representation sharing, this work extends it by preserving auxiliary outputs beyond training as an inference-time explainability and decision-control interface.

Contributions:

- A bounded three-class decision formulation (YES/NO/TBD) that treats deferral as a first-class trained output, not a confidence-threshold artifact.
- A multi-task architecture that retains auxiliary explainability outputs (Human Values, Emotions/Sentiments) at inference time as structured decision-policy control variables.
- An empirical evaluation on 22,748 held-out examples with full per-class breakdown, confusion matrix, ablation, and latency characterization.

2 System Overview

The system receives a tokenized decision event, evaluates it through a shared transformer encoder, and produces three outputs: a bounded decision ($y \in \{\text{YES}, \text{NO}, \text{TBD}\}$), a structured Human Values vector ($v \in \{0, 1\}^{10}$), and a structured Emotions/Sentiments vector ($e \in \{0, 1\}^{28}$). All three outputs are available at inference time. The auxiliary outputs serve as explainability signals and control variables for downstream reasoning, thresholding, and reason-code generation—they are not discarded after training.

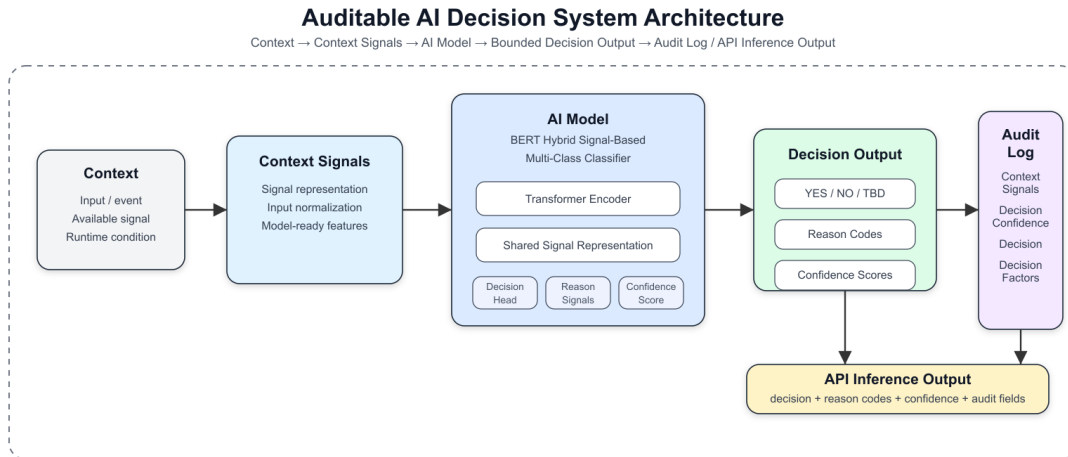


Figure 1: System architecture: tokenized decision event → shared transformer encoder with auxiliary heads → bounded decision output with structured explainability outputs (Human Values, Emotions/Sentiments) and confidence score.

The full model jointly predicts $f(x) = (f_d(x), f_v(x), f_e(x))$, where f_d is the bounded decision function

and f_v , f_e are the auxiliary context functions retained for inference-time control.

3 Methodology

3.1 Architecture

The model uses a transformer-based multi-task architecture initialized from bert-base-uncased [2]. A shared encoder processes tokenized inputs (maximum 128 tokens) and produces a contextual representation consumed by three task-specific output heads:

- Decision head: three-class classifier producing $y \in \{\text{YES}, \text{NO}, \text{TBD}\}$.
- Value head: 10-label multi-label predictor for Human Values.
- Emotion head: 28-label multi-label predictor for Emotions/Sentiments.

Confidence scores are derived from the primary decision head softmax and exposed at inference alongside the auxiliary outputs. Calibration of neural classifier confidence is a known challenge [6]; the TBD class provides a native low-confidence route independent of softmax calibration.

3.2 Training

The model is trained jointly on the primary bounded decision task and the two auxiliary multi-label tasks under the following configuration:

- Backbone: bert-base-uncased; maximum sequence length: 128 tokens
- Batch size: 32
- Learning rate: layer-wise differential rates across encoder layers and task-specific heads; cosine decay with warmup ratio 0.1
- Random seed: 42
- Class imbalance handling: per-class decision weights applied to the primary head loss

This joint formulation ensures that decision outputs, auxiliary signals, and confidence scores are learned from a shared representation, enabling consistent behavior across prediction, explanation, and downstream control.

4 Results and Analysis

4.1 Dataset and Evaluation Setup

The corpus contains up to 181,000 labeled decision events. Each example carries a primary bounded decision label (YES/NO/TBD) and multi-label annotations for Human Values (10 labels) and Emotions/Sentiments (28 labels). Results are reported on a stratified held-out test split of 22,748 examples:

Class	Count	Share
YES	5,871	25.8%
NO	3,159	13.9%
TBD	13,718	60.3%
Total	22,748	100.0%

TBD is the majority class (60.3%), so Macro F1 is the primary evaluation metric.

4.2 Performance Comparison

Method	Accuracy	Macro F1	Micro F1	Weighted F1
Majority-action baseline (always TBD)	0.6030	0.2508	0.6030	0.4537
Bounded-decision ablation	–	0.8211	–	–
Proposed system	0.8485	0.8215	0.8485	0.8506

4.3 Per-Class Performance

Class	Precision	Recall	F1	Support
YES	0.7683	0.9029	0.8302	5,871
NO	0.7164	0.7923	0.7524	3,159
TBD	0.9306	0.8381	0.8819	13,718

4.4 Confusion Matrix

	Pred YES	Pred NO	Pred TBD
True YES	5,301	141	429
True NO	228	2,503	428
True TBD	1,371	850	11,497

4.5 Inference Latency

Measured over 200 inference runs on NVIDIA Tesla T4 (16 GB): p50 = 200 ms, p95 = 415 ms.

5 Discussion

The 0.5707 Macro F1 improvement over the majority-action baseline confirms that the model learns genuine three-way discrimination rather than exploiting class imbalance. The TBD class constitutes 60.3% of the test split; a system predicting TBD exclusively achieves 60.3% accuracy. The proposed system reaches 84.9% accuracy with strong coverage across all three decision states.

The ablation result (0.0019 Macro F1 gap) shows that the auxiliary channels are not the primary source of classification performance. Their operational value is at inference time: the Human Values and Emotions/Sentiments outputs allow downstream systems to inspect decision-policy context, apply deployment-specific thresholds, and generate structured reason codes without modifying the core model.

The dominant error mode—true TBD predicted as YES—highlights the boundary between action and deferral. This boundary can be tuned at deployment time using confidence scores and auxiliary signals, providing a practical mechanism for risk-sensitive decision control.

More broadly, this formulation reframes classification from a forced labeling problem into a decision interface problem, where abstention, explanation, and confidence are co-equal outputs. This shift is particularly relevant for systems deployed under uncertainty, where the cost of incorrect action exceeds the cost of deferral.

6 Limitations

Results are specific to the dataset, label spaces, and evaluation environment used in this study. While the decision formulation and signal-based representation are designed to be domain-agnostic, empirical validation across additional decision domains remains an area for future work. Class imbalance in the NO class (13.9% of the test split) likely constrains NO performance and may require additional data or targeted loss shaping. Reported latency figures are hardware-dependent. Fuller isolation of individual auxiliary channel contributions to bounded decision behavior is deferred to future work.

7 Conclusion

This paper presented an auditable AI decision system with bounded decisions (YES/NO/TBD), structured explainability through auxiliary Human Values and Emotions/Sentiments outputs, and per-decision confidence scores. On a stratified held-out test split of 22,748 examples from a corpus of up to 181,000 labeled decision events, the system achieves Macro F1 = 0.8215 and Accuracy = 0.8485, outperforming a majority-action baseline by 0.5707 Macro F1 points. Per-class F1 scores are 0.8302 (YES), 0.7524 (NO), and 0.8819 (TBD). An ablation confirms that auxiliary channels contribute inference-time control variables rather than classification gain. Inference latency is $p50 = 200$ ms and $p95 = 415$ ms on an NVIDIA Tesla T4 GPU.

Beyond performance improvements, this work introduces a shift from forced classification toward decision interfaces that explicitly incorporate deferral, explainability, and confidence as first-class outputs. By separating decision prediction from decision control, the system enables execution-time adaptation without retraining, allowing downstream systems to apply thresholds, policies, and constraints dynamically. The bounded decision formulation ensures that uncertainty is represented explicitly rather than suppressed, while auxiliary signals provide a structured and auditable basis for interpreting and steering decisions. This combination positions the system not merely as a classifier, but as a practical decision control layer for real-world AI deployments operating under uncertainty.

References

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